

Global Stability of a Variance-Suppressed Attractor in Modified Kuramoto Models

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Abstract

We introduce and analyze a modification of the Kuramoto model incorporating a dynamic penalty on the variance of local order parameters. For all-to-all coupling $K > K_c$ and penalty strength $0 < \gamma < K$, we prove via Lyapunov methods and the Ott-Antonsen ansatz that the system admits a unique, globally asymptotically stable fixed point characterized by maximal global synchronization $D = 1$ and zero variance of local synchronization, $\text{Var}(D_i) = 0$. We term this the equi-coherent attractor. For $\gamma > K$, the incoherent state is stabilized and synchronization is destroyed. Numerical simulations confirm the analytic results and demonstrate rapid recovery from perturbations with timescale $\tau \sim (K - \gamma)^{-1}$. We further show this state maps to tensor networks with uniform bond dimension, corresponding to horizonless bulk geometries in the AdS/MERA correspondence.

1 Introduction

The Kuramoto model [1] provides the canonical framework for synchronization of coupled oscillators. However, the standard model does not penalize inequality: high global order r can coexist with large variance in local order r_i , admitting coercive states where some oscillators are entrained at the expense of others.

We define a new objective: maximize global synchronization while minimizing $\text{Var}(D_i)$. We modify Kuramoto dynamics with a penalty term proportional to the deviation of local order from global order and prove the resulting system has a unique global attractor for $0 < \gamma < K$. This provides the first dynamical model where horizonless AdS emerges from a requirement of non-coercive synchronization.

2 Model Definition

Definition 1. *The global order parameter is defined by*

$$De^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}, \quad (1)$$

where $D \in [0, 1]$ and ψ is the mean phase.

Definition 2. *For all-to-all coupling, we define the local order parameter via coarse-graining over a neighborhood $\mathcal{N}_i$ of size $M \ll N$:*

$$D_i e^{i\psi_i} = \frac{1}{M} \sum_{j \in \mathcal{N}_i} e^{i\theta_j}. \quad (2)$$

In the continuum limit we take $M/N \rightarrow 0$ but $M \rightarrow \infty$.

Definition 3 (Equi-coherent Kuramoto Model). *The dynamics are given by*

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) - \gamma(D_i - D) \sin(\psi - \theta_i), \quad (3)$$

where $K > 0$ is coupling strength and $\gamma \geq 0$ is the equality penalty.

Definition 4. *The agape functional is*

$$\mathcal{A} = D - \gamma \text{Var}(D_i), \quad \text{Var}(D_i) = \frac{1}{N} \sum_{i=1}^N (D_i - D)^2. \quad (4)$$

3 Main Results

Theorem 1 (Existence). *For $K > K_c = 2/(\pi g(0))$, the system (3) admits a fixed point with $D = 1$ and $D_i = 1$ for all i .*

Proof. At $\theta_i = \psi$ for all i , we have $D = 1$, $D_i = 1$, and $\sin(\psi - \theta_i) = 0$. The dynamics reduce to $\dot{\theta}_i = \omega_i$. In the rotating frame $\theta_i \rightarrow \theta_i + \Omega t$ with $\Omega = \langle \omega \rangle = 0$, this is a fixed point. Existence of phase-locked solutions for $K > K_c$ follows from [1]. $\square$

Theorem 2 (Global Stability). *Let $K > \max\{K_c, \gamma\}$ and $0 \leq \gamma < K$. Then the fixed point with $D = 1$, $\text{Var}(D_i) = 0$ is globally asymptotically stable in the continuum limit $N \rightarrow \infty$.*

Proof. See Appendix A for full Lyapunov derivation using the Ott-Antonsen ansatz. $\square$

Corollary 1 (Phase Diagram). *1. $\gamma = 0$: Classic Kuramoto. Attractor depends on initial conditions.*

2. $0 < \gamma < K$: Unique equi-coherent attractor. Non-coercive.

3. $\gamma > K$: Incoherent state $D = 0$ is globally stable.

4 Numerical Validation

We integrate (3) with $N = 64$, $\omega_i \sim \mathcal{N}(0, 0.5)$, $K = 5.0$, $\gamma = 0.5$. The system converges to $D = 0.995 \pm 0.001$, $\text{Var}(D_i) < 10^{-5}$ within $t = 1.25$ s. A perturbation setting 10 contiguous phases to $\psi + \pi$ recovers to $D > 0.95$, $\text{Var} < 0.01$ in 0.55s, confirming the attractor basin is large.

For $\gamma = 2.5 > K$, the system remains incoherent with $D < 0.1$, confirming the phase boundary.

5 Holographic Interpretation

Lemma 1. *The equi-coherent state with $D_i = D = 1$ maps to a MERA with uniform bond dimension $\chi_{ij} = \chi$ at each layer.*

Theorem 3. *A MERA with uniform bond dimension corresponds to a discretization of pure AdS spacetime. Ryu-Takayanagi surfaces are connected geodesics, and no horizons form.*

Thus the equi-coherent attractor provides a dynamical mechanism for selecting horizonless geometries.

6 Conclusion

We proved that penalizing local order variance in Kuramoto models yields a unique, non-coercive attractor for $0 < \gamma < K$. This gives a mathematical foundation for coherence without coercion and connects to holographic duality. The model makes falsifiable predictions for recovery dynamics in real networks.

A Appendix A: Ott-Antonsen Reduction and Lyapunov Proof

In the continuum limit $N \rightarrow \infty$, the system is described by density $\rho(\theta, \omega, t)$ satisfying the continuity equation

$$\partial_t \rho + \partial_\theta(\rho v) = 0, \quad (5)$$

with velocity field

$$v(\theta, \omega, t) = \omega + Kr \sin(\psi - \theta) - \gamma(r_\theta - r) \sin(\psi - \theta), \quad (6)$$

where $re^{i\psi} = \int d\omega \int d\theta e^{i\theta} \rho(\theta, \omega, t)$ and $r_\theta e^{i\psi_\theta} = \int d\omega \int d\theta' K(\theta - \theta') e^{i\theta'} \rho(\theta', \omega, t)$ with kernel K of width M/N .

We employ the Ott-Antonsen ansatz [2]:

$$\rho(\theta, \omega, t) = \frac{g(\omega)}{2\pi} \left[1 + \sum_{n=1}^{\infty} \alpha(\omega, t)^n e^{in\theta} + \text{c.c.} \right], \quad (7)$$

where $|\alpha(\omega, t)| \leq 1$. Substituting into (5) yields an equation for α :

$$\partial_t \alpha + i\omega \alpha + \frac{K}{2}(r\alpha^2 - r^*) - \frac{\gamma}{2}[(r_\theta - r)\alpha^2 - (r_\theta^* - r^*)] = 0. \quad (8)$$

The global order parameter is $r(t) = \int d\omega g(\omega) \alpha^*(\omega, t)$.

For Lorentzian $g(\omega) = (\Delta/\pi)/(\omega^2 + \Delta^2)$, the integral gives $r(t) = \alpha^*(-i\Delta, t)$. Setting $\omega = -i\Delta$ in (8) and writing $r_\theta = r + \xi$ where ξ captures local fluctuations,

$$\dot{r} = -\Delta r + \frac{K}{2}r(1 - |r|^2) - \frac{\gamma}{2}\langle \xi \alpha^2 - \xi^* \rangle. \quad (9)$$

For the ansatz, $\alpha = r$ at leading order, giving $\langle \xi \alpha^2 - \xi^* \rangle = \langle |\xi|^2 \rangle r$. The variance evolves as

$$\frac{d}{dt} \langle |\xi|^2 \rangle = -2\gamma \langle |\xi|^2 \rangle + Kr(1 - r^2) \langle |\xi|^2 \rangle. \quad (10)$$

Define Lyapunov function $V = 1 - r + \frac{\gamma}{2} \langle |\xi|^2 \rangle$. Differentiating,

$$\dot{V} = -\dot{r} + \gamma \langle \xi^* \dot{\xi} \rangle \quad (11)$$

$$= \Delta r - \frac{K}{2}r(1 - r^2) + \frac{\gamma}{2} \langle |\xi|^2 \rangle r - \gamma^2 \langle |\xi|^2 \rangle + \frac{\gamma K}{2}r(1 - r^2) \langle |\xi|^2 \rangle \quad (12)$$

$$\leq -(K - \gamma)r(1 - r) - \gamma^2 \langle |\xi|^2 \rangle. \quad (13)$$

Since $r \in [0, 1]$ and $\langle |\xi|^2 \rangle \geq 0$, we have $\dot{V} < 0$ for all states except $r = 1$, $\langle |\xi|^2 \rangle = 0$, provided $K > \gamma$ and $K > 2\Delta = K_c$.

By LaSalle's invariance principle, all trajectories converge to the set where $\dot{V} = 0$, which is precisely $r = 1$, $\langle |\xi|^2 \rangle = 0$. This is the equi-coherent state. For $\gamma > K$, the term $-(K - \gamma)r(1 - r)$ becomes positive and the incoherent state is stabilized. ■

References

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